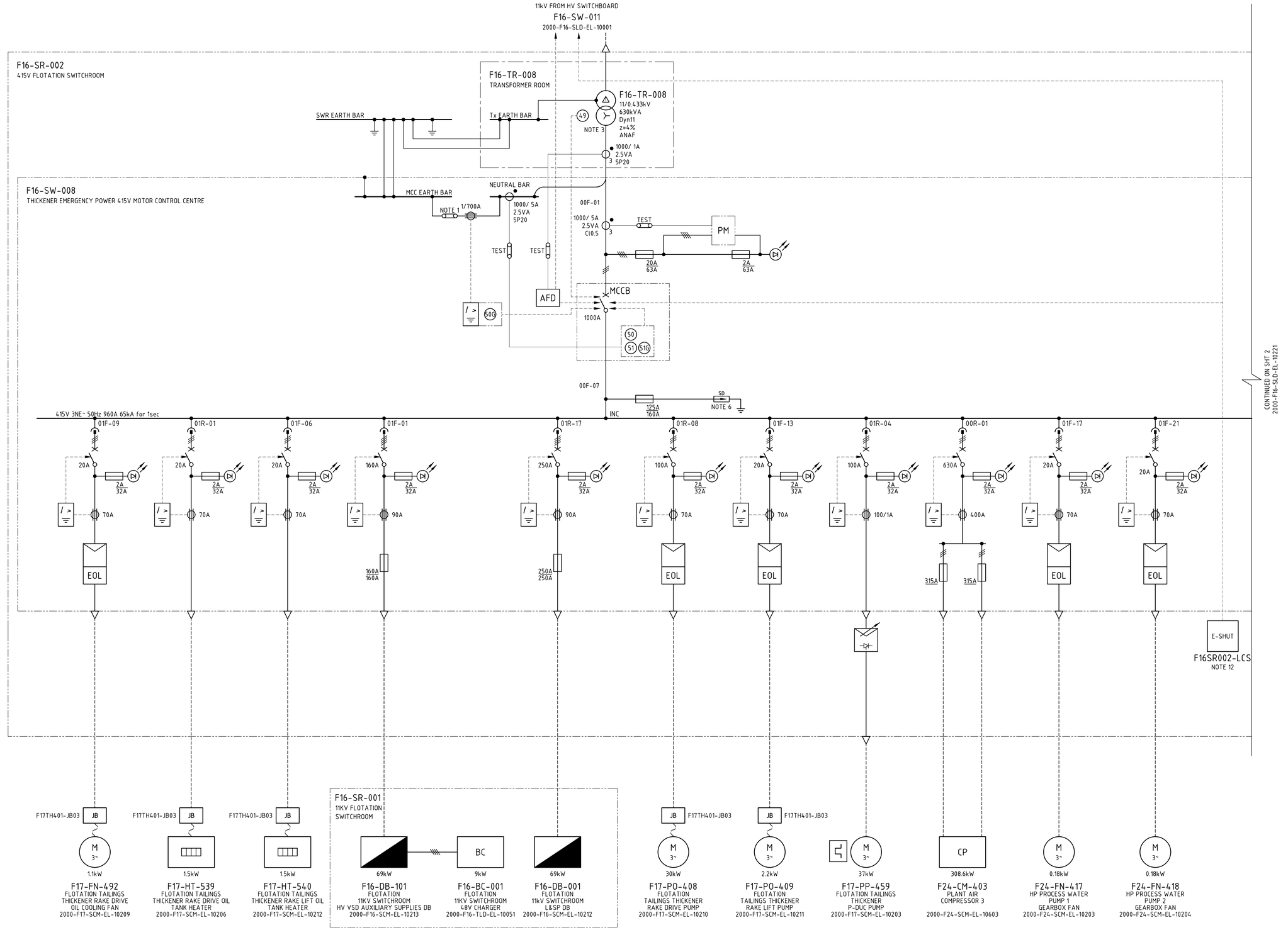


NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F16-SW-008 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F16-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- BIMETALLIC TEMPERATURE SWITCH
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES	REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2000-F16-SCM-EL-10273	F16-SW-008 ARC FLASH SYSTEM SCHEMATIC DIAGRAM								
2000-F16-SCM-EL-10272	F16-SW-008 24VDC DISTRIBUTION SCHEMATIC DIAGRAM								
2000-F16-SCM-EL-10270	F16-SW-008 INCOMER SCHEMATIC DIAGRAM SHEET 1								
2000-F16-DRG-EL-10101	F16-SR-002 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT								
2000-F16-BLK-IC-10001	F16-SR-002 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM	3	07SEP26	KCGM GROWTH PROJECT - SP1 AS-BUILT					
2000-F16-SLD-EL-10222	F16-SW-008 415V EMERGENCY MCC SINGLE LINE DIAGRAM SHEET 2	2	26AUG25	FAT REDLINES INCORPORATED					
2000-F16-SLD-EL-10001	F16-SW-011 11kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1	1	05JUL24	INCOMER CHANGED FROM ACB TO MCCB, BUS RATING WAS 1000A					
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX	0	11JUN24	APPROVED FOR CONSTRUCTION					

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: ROUGHER AND SCAVENGER FLOTATION
F16-SW-008 415V EMERGENCY MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F16-SLD-EL-10221	3

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235I7 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SLD-EL-10221.DWG 03SEP26 2:46PM ROBINLAW